CHAPTER 9

Possibilities, Preferences, and Choices

ECON101 - Prince Sultan University

→ PRACTICE CALC/GRAPHS
so exam will be easy
ch8+ch9 - ch6 just review

ch8- calculations
graphs → ch6, ch9
+ calc

2 questions theory
(1 mark easy)

OVERVIEW

Learning objectives

graphs
calculations

A household's purchases depend on its **income**, the **prices** it faces, and its **preferences**. A **budget line** shows what the household can afford; **indifference curves** describe its preferences. Together, they predict how choices change when prices or income change.

rotates

1. Construct a household's budget line and explain how it moves when prices or income change. → shifts
2. Represent preferences with indifference curves and explain why the marginal rate of substitution diminishes. × didn't cover yet
3. Predict how a household's consumption choices respond to changes in prices and income.

OVERVIEW

Running example: Rakan's monthly budget

Rakan divides his whole monthly budget between two goods: swimming-pool sessions and fruit.

144 SAR
Income to spend each month (Y)

18 SAR
Price of a swimming session (P_S)

12 SAR
Price of fruit per kg (P_F)

Later sections change the price of a session and then Rakan's income, and trace how his choices adjust.

construct the table with the help of the budget equation

CONSUMPTION POSSIBILITIES

The budget line

boundary that separates combinations a household can afford from those that it cannot

The **budget line** separates the combinations a household can afford from those it cannot.

Spending all 144 SAR, Rakan can buy:

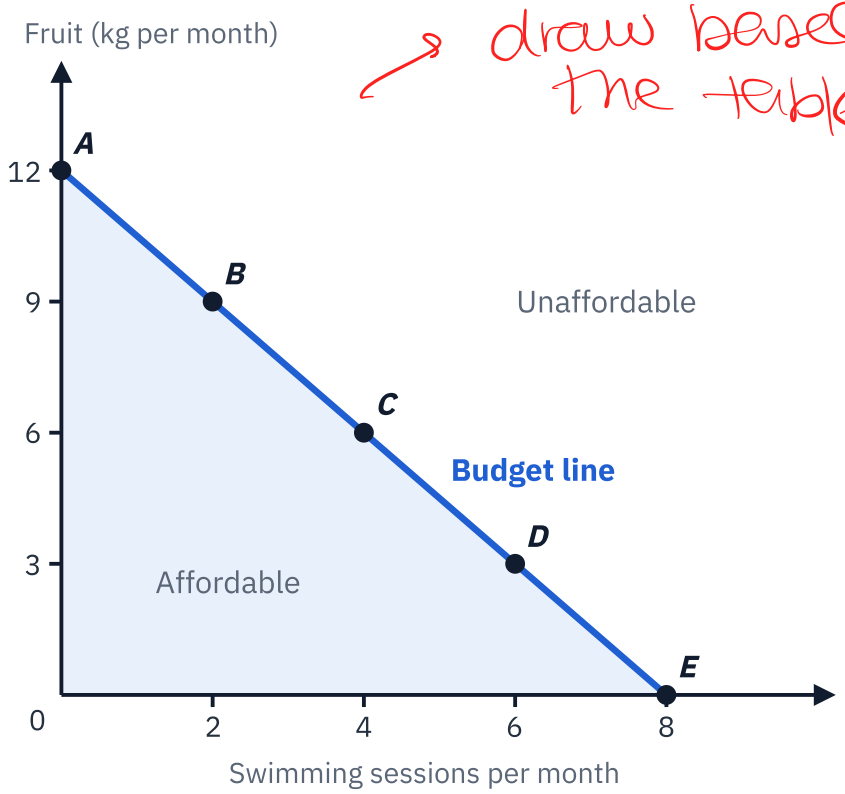
exhausting entire budget

$\frac{144}{18}$

Possibility	Swimming sessions (18 SAR)		Fruit (kg) (12 SAR)		Spending (SAR)
A	$0 \times 18 = 0$	0	$12 \times 12 = 144$	12	0 + 144 = 144
B	$2 \times 18 = 36$	2	$9 \times 12 = 108$	9	36 + 108 = 144
C	$4 \times 18 = 72$	4	$6 \times 12 = 72$	6	72 + 72 = 144
D	$6 \times 18 = 108$	6	$3 \times 12 = 36$	3	108 + 36 = 144
E	$8 \times 18 = 144$	8	$0 \times 12 = 0$	0	144 + 0 = 144

* ask how to construct table under tight time

Rakan's budget line



- A swimming session is an **indivisible good**: it is sold only in whole units. Fruit is a **divisible good**: any amount can be bought.
- Rakan's budget line runs through points *A* to *E*.
- The budget line **constrains** Rakan's choices. Any combination **on or inside** the line with a whole number of sessions is affordable (3 sessions and 7.5 kg, for example); combinations beyond it are not.

Price$_S$ Quantity$_S$ + Price$_F$ Quantity$_F$ = Y

CONSUMPTION POSSIBILITIES

The budget equation

The budget line can also be written as a **budget equation**, which sets spending equal to income: **expenditure = income**. Let P_S and Q_S be the price and quantity of sessions, P_F and Q_F the price and quantity of fruit, and Y income:

$$P_S Q_S + P_F Q_F = Y \Rightarrow 18 Q_S + 12 Q_F = 144$$

18 12

Dividing through by P_F and subtracting $(P_S/P_F)Q_S$ from both sides gives fruit in terms of sessions:

$$Q_F = Y/P_F - (P_S/P_F)\, Q_S \Rightarrow Q_F = 12 - 1.5\, Q_S$$

12 18/12

fruits in terms of sessions

2nd is on y-axis?

one item foregone

fruit in terms of swimming

CONSUMPTION POSSIBILITIES

Real income and relative price

Real income → income measured in terms of goods

Real income is income measured in goods: the quantity of a good that a household's income will buy.

In terms of fruit, Rakan's real income is $Y/P_F = 144/12 = \mathbf{12\ kg}$, the budget line's intercept on the y-axis.

Relative price

MRS = relative price → satisfaction in indifference curve

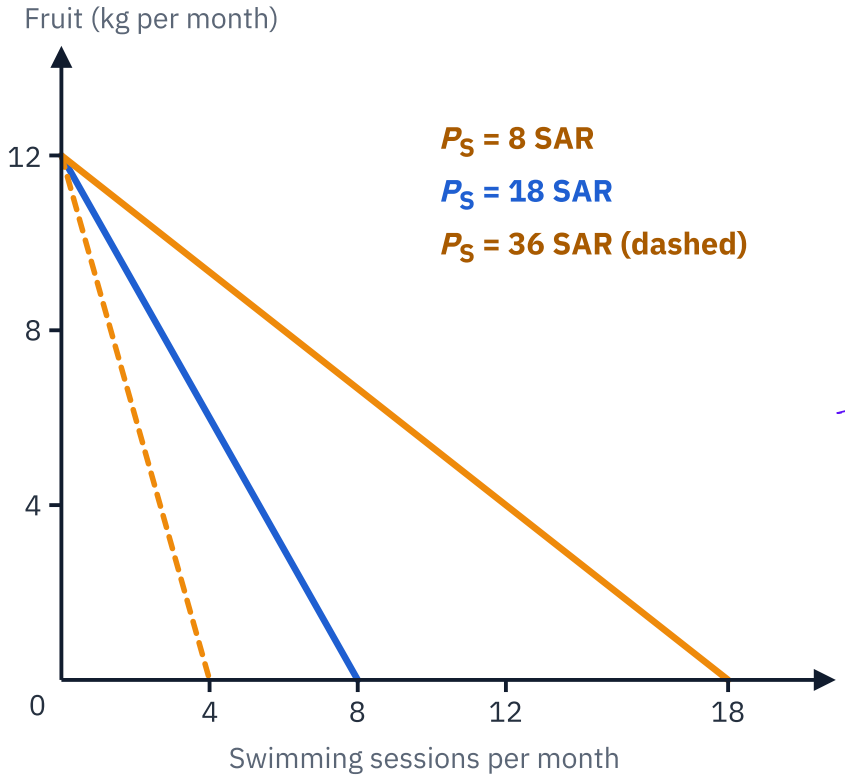
A **relative price** is one good's price divided by another good's price.

The relative price of a session in terms of fruit is $P_S/P_F = 18/12 = \mathbf{1.5}$. The budget line's slope is −1.5: its **magnitude** equals the relative price.

The relative price is Rakan's opportunity cost of a session: each extra session means giving up 1.5 kg of fruit.

$\frac{P_S}{P_F} = \frac{18}{12}$

A change in prices

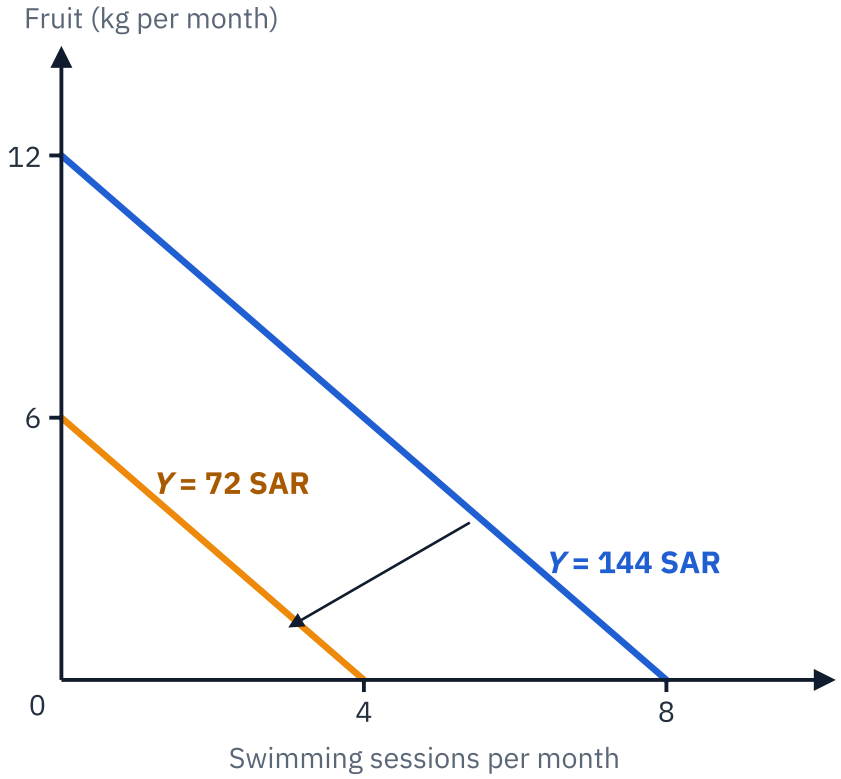


If the price of the good on the horizontal axis changes, the **slope** of the budget line changes. The line pivots around its *y*-axis intercept, which does not move.

- If the price of a session **falls** from 18 SAR to 8 SAR, the relative price falls to 8/12 = 2/3 and the line rotates **outward**: Rakan can afford up to 18 sessions.
- If the price **rises** to 36 SAR, the relative price rises to 3 and the line rotates **inward** (dashed): at most 4 sessions.

CONSUMPTION POSSIBILITIES

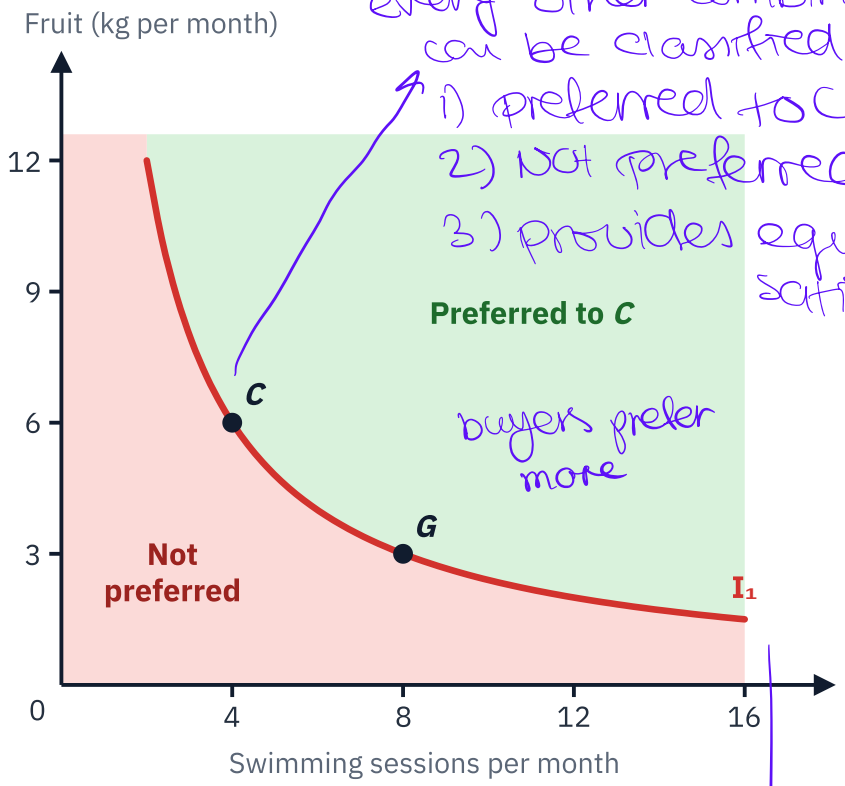
A change in income



When money income changes, the budget line moves **parallel** to itself. Its slope stays the same because relative prices stay the same.

- If Rakan's income **falls** from 144 SAR to 72 SAR, the budget line shifts **leftward**.
- His real income falls to 6 kg of fruit or 4 sessions. The slope stays at −1.5 because the relative price is still 1.5.

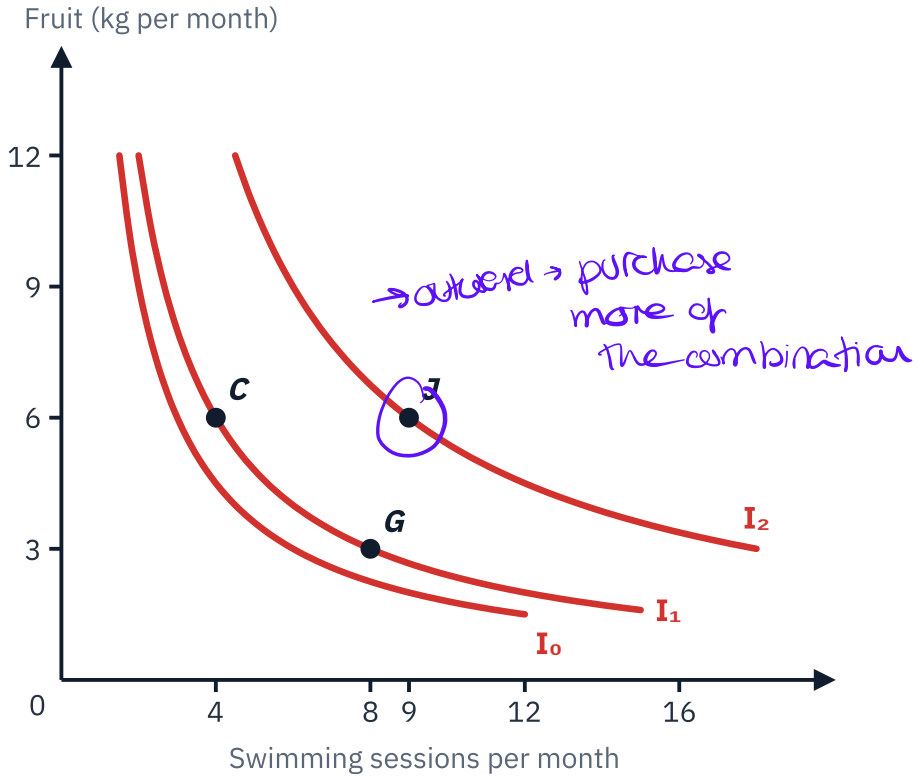
An indifference curve



Rakan's preferences can be pictured as a map built from indifference curves. An **indifference curve** links all the combinations of goods that a consumer likes equally well, so the consumer is **indifferent** among them.

- Start from *C*: **4 sessions and 6 kg** of fruit a month. Rakan can place every other combination in one of three groups: **preferred** to *C*, **not preferred** to *C*, or **just as good as** *C*.
- The curve I_1 passes through the combinations that are just as good, such as *G* (8 sessions and 3 kg).
- Every point **above** I_1 is preferred to every point on it, and every point on I_1 is preferred to every point **below** it.

A preference map



A **preference map** draws several indifference curves together, and points on a higher curve are preferred to points on a lower one.

- Each point on I_2 is preferred to each point on I_1, which in turn is preferred to each point on I_0.
- So Rakan prefers J to both C and G.

PREFERENCES AND INDIFFERENCE CURVES

Marginal rate of substitution

The **marginal rate of substitution** (*MRS*) is the rate at which a person is willing to give up good *y* to get more of good *x* while staying **indifferent**, that is, on the same indifference curve. On a graph, the *MRS* is the steepness of the indifference curve: the **magnitude of its slope**.

Steep indifference curve

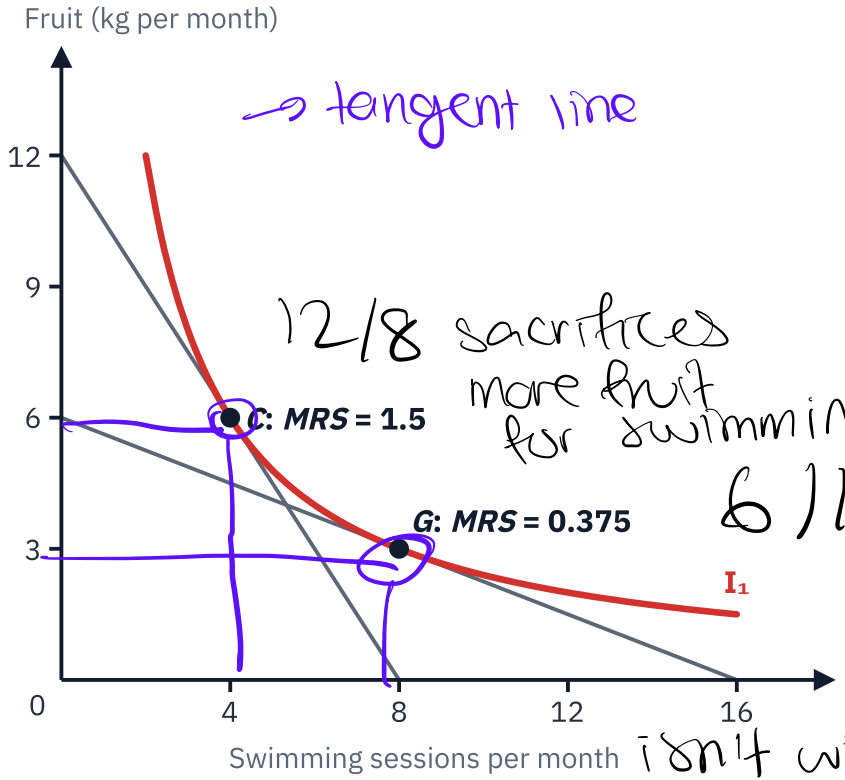
A **steep** curve means a **high** *MRS*: the person would trade a lot of *y* for a little more *x*.

Flat indifference curve

A **flat** curve means a **low** *MRS*: the person would give up only a little *y* for more *x*.

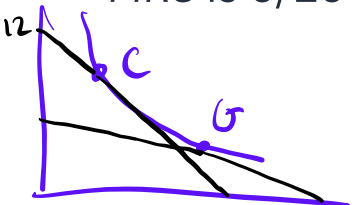
The **key assumption** of consumer theory is a **diminishing marginal rate of substitution**: the more of good *x* a person already has, the less of good *y* they are willing to give up per extra unit of *x* while staying indifferent. As the next slide shows, Rakan's *MRS* falls from 1.5 at *C* to 0.375 at *G*. This assumption is what makes indifference curves **bowed toward the origin**.

Measuring a diminishing MRS



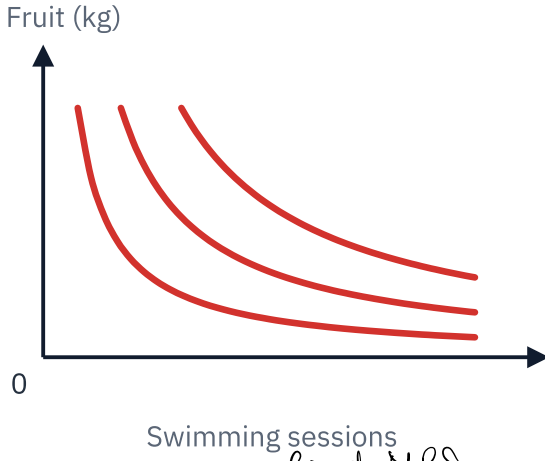
To find the *MRS* at a point, draw the straight line that touches the curve only at that point (the **tangent**) and take the magnitude of its slope:

- At *C* (4 sessions, 6 kg), the tangent falls 12 kg over 8 sessions, so Rakan is willing to trade fruit for sessions at a rate of 12/8 = 1.5 kg per session: his *MRS* is **1.5**.
- At *G* (8 sessions, 3 kg), the tangent falls 6 kg over 16 sessions, so his *MRS* is 6/16 = **0.375**.



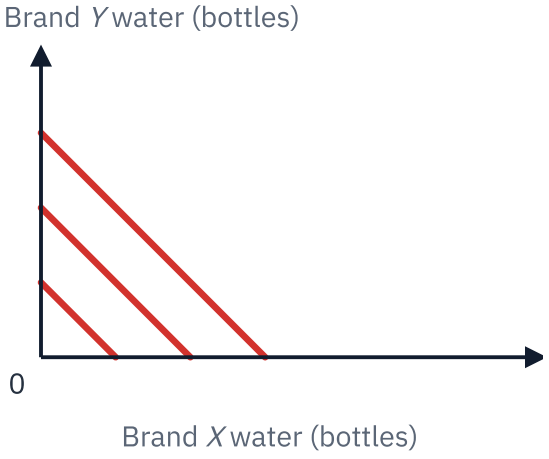
Degree of substitutability

How easily one good can replace another, the **degree of substitutability**, shows up in the shape of the indifference curves.



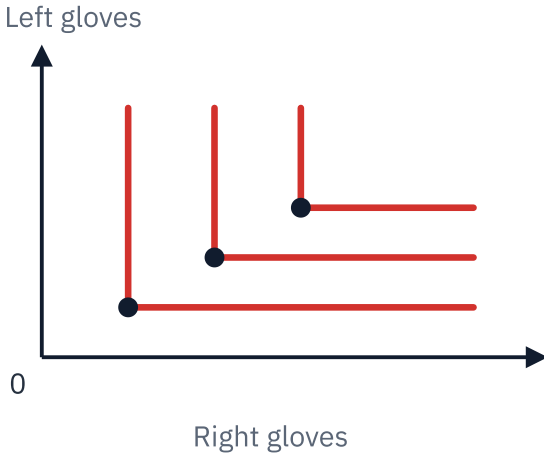
(a) Ordinary goods

The curves bow toward the origin; the *MRS* diminishes. Enough extra sessions can make up for less fruit.



(b) Perfect substitutes

Downward-sloping straight lines; the *MRS* is constant. A bottle of Brand *X* can replace one of Brand *Y* one for one.



(c) Perfect complements

L-shaped curves. An extra right glove is worthless without a left one; two of each is preferred to one of each.

The best affordable choice

For ordinary goods like Rakan's, the **best affordable choice** is the point that:

1

Lies **on the budget line**, so all income is spent.

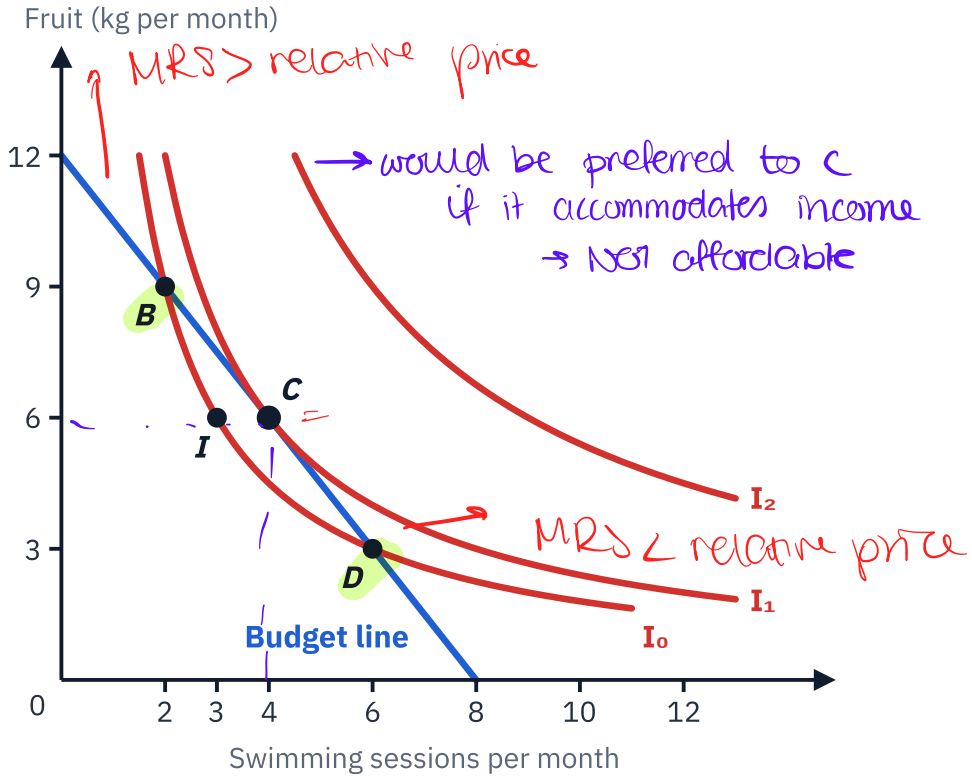
2

Reaches the **highest indifference curve** the budget allows.

3

Is where the **marginal rate of substitution equals the relative price**.

Rakan's best affordable choice



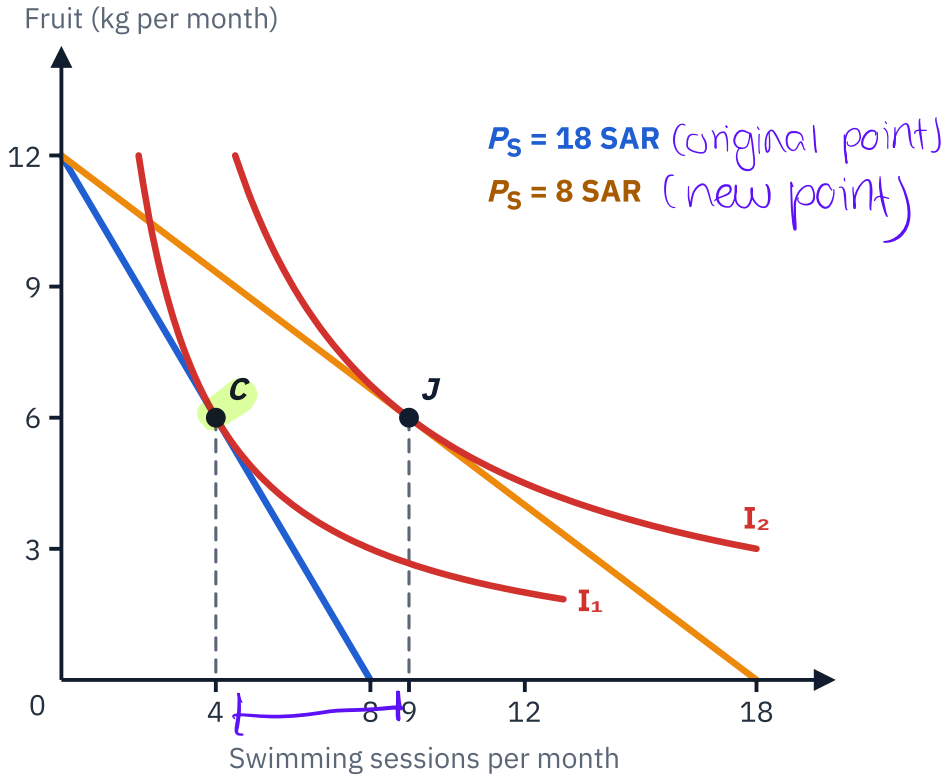
- Rakan's best affordable point is ***C*:** **4 sessions and 6 kg** of fruit a month.
- *B* (more fruit, fewer sessions) and *D* (more sessions, less fruit) are also affordable. But both lie on I_0, the same indifference curve as *I*, and *C* is preferred to *I*. Points on I_2 would be better than *C* but cost more than 144 SAR.
- At *B*, I_0 is steeper than the budget line: *MRS* > relative price. At *D*, I_0 is flatter: *MRS* < relative price.

At *C*, I_1 just touches the budget line: *MRS* = P_S/P_F = 1.5.

16 / 25

MRS = relative price

A change in price: the price effect



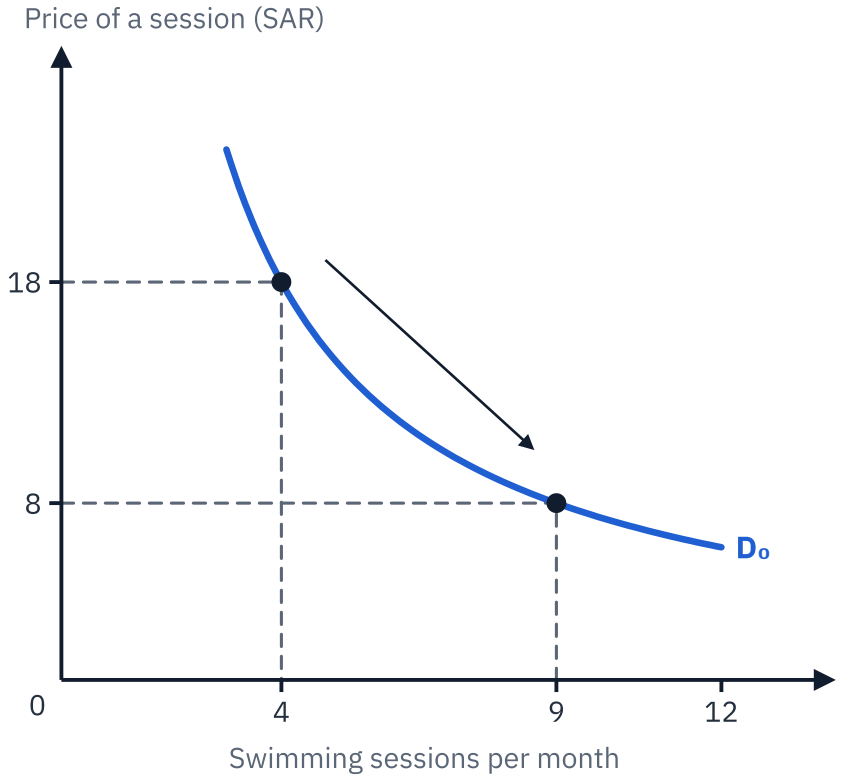
The **price effect** is the change in the quantity of a good consumed that results from a change in its price, with income and other prices unchanged.

At first a session costs 18 SAR and Rakan is at *C*. When the price **falls to 8 SAR**, his budget line rotates outward and his best affordable point becomes ***J*: 9 sessions and 6 kg of fruit**.

The lower price raises the number of sessions Rakan buys from 4 to 9 a month. → +5

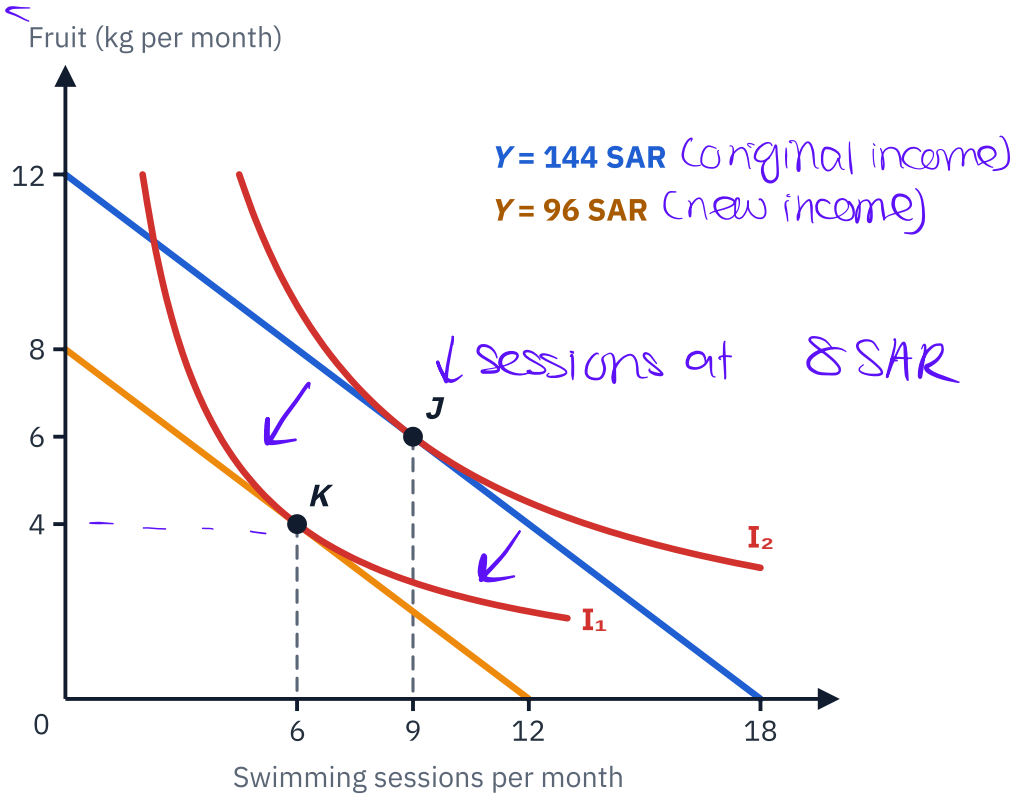
lower price increases session

The price effect and the demand curve



- The change from 4 sessions at 18 SAR to 9 sessions at 8 SAR is a **movement along** Rakan's demand curve for swimming sessions.
- The marked points are best affordable choices; the smooth curve through them illustrates the demand curve D_0.

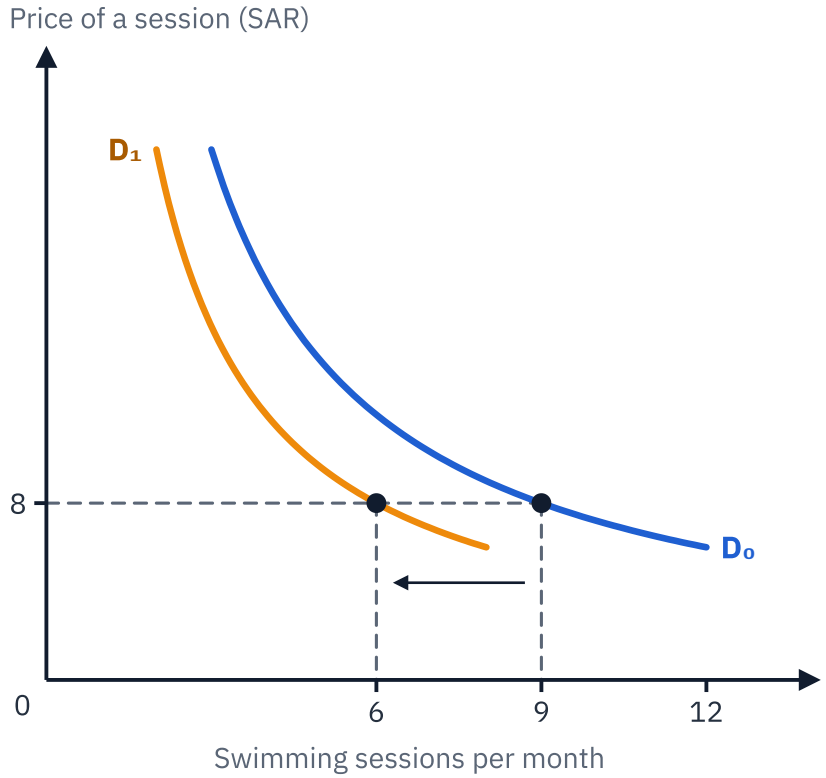
A change in income: the income effect



The **income effect** is the change in buying plans caused by a change in income, with prices held constant.

With sessions at 8 SAR, Rakan is at *J* (9 sessions). His income then **falls from 144 SAR to 96 SAR** while prices are unchanged. The budget line shifts leftward and his best affordable point becomes ***K*: 6 sessions and 4 kg of fruit**.

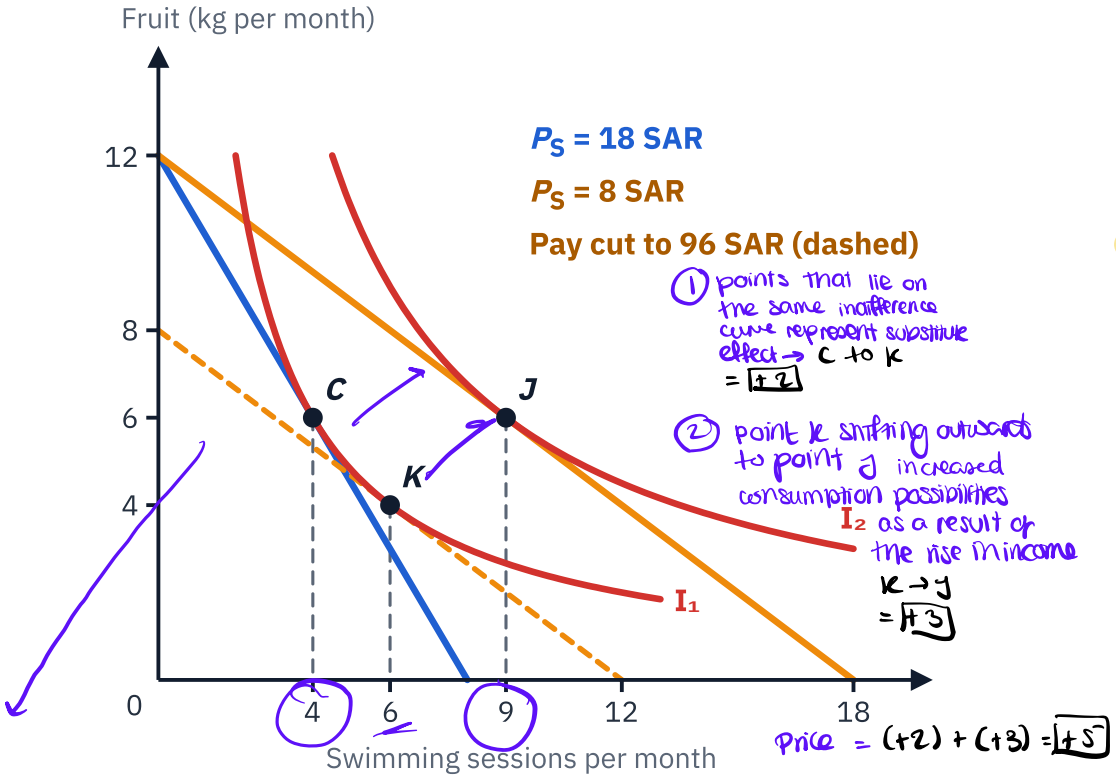
The income effect and demand



- Rakan's **demand for swimming sessions decreases**. At 8 SAR a session he now buys 6 sessions, not 9, so his demand curve **shifts leftward** from D_0 (income 144 SAR) to D_1 (income 96 SAR).
- The lower income reduces his purchases of both goods, so for Rakan swimming sessions and fruit are **normal goods**.

→ if demand shifts rightward then it's identified as an inferior good

Substitution effect and income effect



When the price of a normal good falls, the quantity consumed **always** rises. The reason becomes clear once the price effect is split into two parts. Rakan's income is 144 SAR and the price of a session falls from 18 SAR to 8 SAR.

Substitution effect: *C* → *K*, 4 → 6 sessions (+2)

Income effect: *K* → *J*, 6 → 9 sessions (+3)

Price effect: *C* → *J*, 4 → 9 sessions (+5)

Price effect (+5) = substitution effect (+2) + income effect (+3).

PREDICTING CONSUMER CHOICES

Isolating the two effects

Substitution effect

The response of the quantity bought to a price change when the consumer is kept on the **same indifference curve**.

- Give Rakan a **hypothetical pay cut** of 48 SAR (to 96 SAR), so that at the new price he is back on his original indifference curve I_1 (dashed budget line). His best affordable point is K.
- The move from C to K, from **4 to 6 sessions**, is the substitution effect.
- This effect always points the same way: a lower relative price leads the consumer to replace other goods with **more** of the cheaper good. It is the first reason demand curves slope downward.

Income effect

- Now cancel the pay cut so that Rakan's income is 144 SAR again. He moves from K on I_1 to J on I_2.
- The move from K to J, from **6 to 9 sessions**, is the income effect. Because swimming is a normal good, the higher income leads him to swim more: the income effect is positive.
- It **adds to** the substitution effect and is the second reason demand curves slope downward.

Inferior goods

An **inferior good** is one that people buy **less** of when their income rises.

- A fall in its price therefore has a **negative** income effect, which pulls **against** the substitution effect.
- If the substitution effect is the larger of the two, the demand curve still slopes downward.
- If the negative income effect were larger than the substitution effect, a price cut would **reduce** the quantity demanded and the demand curve would slope **upward**. This case is extremely rare.

When the price of a good falls	Substitution effect	Income effect	Demand curve
Normal good	Buy more	Buy more	Slopes downward
Inferior good	Buy more	Buy less	Slopes downward if the substitution effect is larger

↓ opposes with income

larger than income

↓ but If income dominates slope upward

→ extremely rare case

Rakan's choices and the facts

Situation	P_S (SAR)	Y (SAR)	Point	Sessions	Fruit (kg)
Starting point	18	144	C	4	6
Session price falls	8	144	J	9	6
Income falls	8	96	K	6	4

The price of fruit stays at 12 SAR per kg. The last row is also the hypothetical pay cut. In each case, Rakan spends all his income and his *MRS* equals the relative price.

Back to the facts

- Observed spending patterns are the outcome of households' best affordable choices.
- When prices or incomes change, the best affordable point moves, and spending patterns move with it.

SOURCE

Source

Michael Parkin (2022) *Microeconomics, Global Edition*, 14th Edition, published by Pearson Education.